ME108 Project 2 - Written Report

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Introduction

For our project, we are studying a 6-inch C-clamp such as the one shown in Figure 1. C-clamps are very commonly used tools in both personal and commercial fabrication settings. Their purpose is to secure a metal, wood, or any other material piece in place to aid in manufacturing processes such as sawing, welding, gluing, and much more. Tightening the threaded screw applies a controlled compressive force to the material, which clamps the piece in place. While the C-clamp is a multipart system, we will just be studying one specific part of the C-clamp, which we are calling the body. We will be excluding the threaded screw in our study. The body is shown in green in Figure 2, and the threaded screw is shown in red.



Figure 1: C-clamp



Figure 2: Part of multipart system we are studying

The C-clamp is an interesting object to study since it is able to withstand very large compressive forces. From experience, C-clamps are sometimes tightened with as much force as the operator can physically push, so we are curious about the stress profile within the body of the C-clamp. By figuring out the stress profile, we can figure out where material can be removed and can then structurally optimize the C-clamp. C-clamps are often quite heavy and have a very robust body, which makes them ideal for structural optimization as well. The main benefit of structural optimization for our study is that a reduced mass can improve ergonomics and reduce material cost, which can reduce manufacturing costs.

With this study, we are trying to accomplish reducing the mass of the body of the C-clamp while ensuring that it still holds up to the original factor of safety. In order to accomplish this, our objective is to minimize the mass of the body of the C-clamp. Our constraint to satisfy is that our optimized C-clamp must have the same factor of safety as a standard 6-inch C-clamp specified below. In other words, the maximum stress from our FEA study of the reduced mass C-clamp must be less than or equal to the maximum stress of the original C-clamp since they are made of the same material with the same yield strength.

It should be noted that, based on feedback we received via Canvas from an instructor and feedback we received from our peers on presenting in class, our study setup and parameters did not need to be changed.

Background

From our literature review, we identified several prior studies analyzing the C-clamp. One such source is a SolidWorks tutorial that includes modeling, performing a FEA study, and analyzing the results for a C-clamp [1]. For our analysis, we will reference the loading methodology and constraints presented in the tutorial, specifically the strategy used to emulate the loading from the screw onto the C-clamp threads.

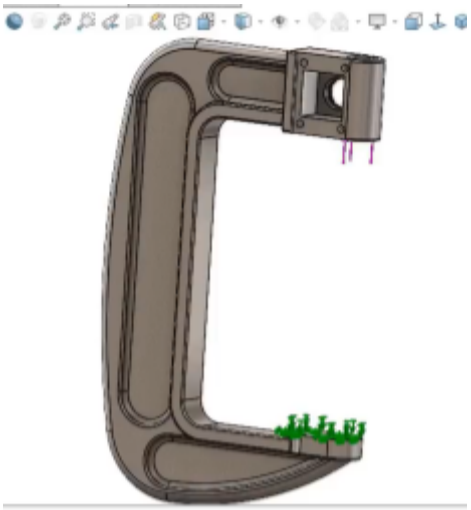


Figure 3: SolidWorks FEA Tutorial Reference

To simplify the simulation, we plan to use the same strategy that the SolidWorks tutorial used of not to include their threads in the model when conducting the FEA analysis. Adding threads will complicate the simulation and the mesh, and is not needed to achieve an accurate result. Instead, the threaded region will be represented by a cylindrical cut through the hole, and our loading force will be applied tangentially to the walls of the cut. This will be shown in more detail below. Additionally, we will use the same fixed constraint as the SolidWorks tutorial, which fixes the contact plane between the clamp and the clamped object and is shown in Figure 3. Overall, this SolidWorks tutorial YouTube video has been a helpful reference to influence the location of our fixed constraint and loading forces for our FEA simulation.

Another study we found was through the ME-738 Computer-Aided Engineering course at New Jersey Institute of Technology. This course includes a paper documenting the entire Elasto-Plastic Analysis of a C-clamp through Ansys [2]. More specifically, this paper runs an FEA simulation on half of a clamp to see how the C-clamp plastically deforms. For our study, we plan to use the mesh strategies that this paper employs. This will allow us to have an accurate reference for our mesh and provide us with one more study to compare our results to.

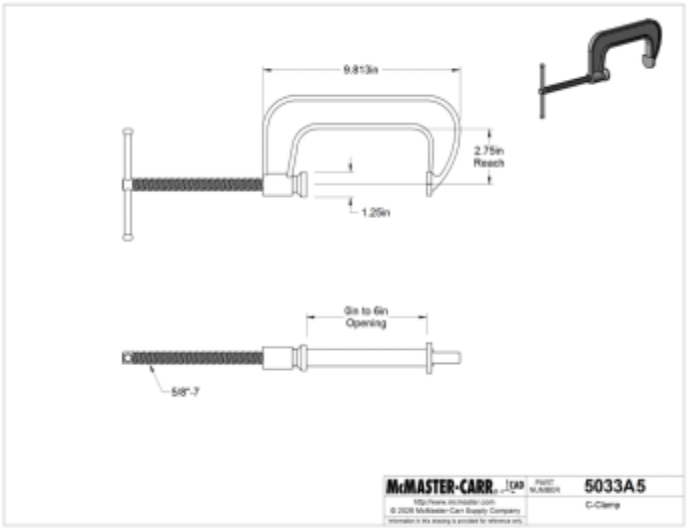


Figure 4: McMaster-Carr 6-in C-Clamp

One final reference for us to establish is a physical C-clamp, which will guide the material, maximum load rating, and model we use. For this, we have decided to choose a commercially available ductile iron C-clamp from McMaster-Carr [3]. Specifically, we will use part number 5033A5, which is seen in Figure 4. The description of this product lists the maximum holding capacity as 2400lb. This geometric model provides a realistic and standardized baseline geometry, where we can also use the rated load as a reference value when performing our FEA and optimization studies.

Method

To set up this study, the first step is to choose a material for our C-clamp. To decide this, we first looked at the body material of our McMaster-Carr reference clamp, which we found to be ductile cast iron. To confirm this material, we also conducted research online through different companies' websites that offer C-clamps and found a common material that they used to also be ductile cast iron. Ductile cast iron is commonly used due to its superior strength, high ductility, and excellent fatigue resistance, which can be seen through its material properties in Figure 5.



Figure 5: Ductile iron material properties

After finding the material to use for our study, we need to determine the load and constraints in our study. Based on McMaster ductile iron 6-inch C-clamp part number 5033A5, we determined earlier that the maximum holding capacity is listed as 2400lb, which is roughly 10,675N. Since we are using the McMaster model for our study, we will use a load of the same magnitude. The direction of this force will be shown later. For our studies, we will apply a fixed constraint on the contact plane between the clamp and the clamped object. This was determined from reference [1] identified above. The visual of this constraint will also be shown later.

After identifying the load and constraint, we need to finalize the initial C-clamp model that we will use for our studies. As mentioned previously, we decided to use the SLDPRT file from the McMaster website for part number 5033A5. However, in order to satisfy the requirements of this project, we need to make some adjustments to the McMaster model. Because a C-clamp is a multipart model, but we are just studying the body of it, we need to first remove the threaded screw and handle from the initial McMaster model. Next, to simplify simulation, we referenced the SolidWorks YouTube tutorial [1] and removed the 5/8 "-7 threads from the model using an extruded cut of a circle with a diameter of 5/8" concentric to the threaded hole on the model. This is shown in Figure 6.

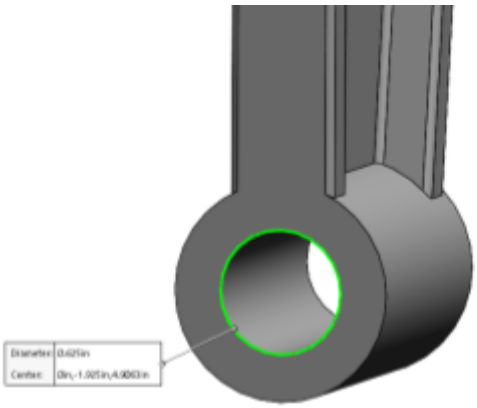


Figure 6: Simplified thread model

Finally, our initial C-clamp model prior to optimization with the extra bodies removed and the thread simplification is shown in Figure 7. This will be used to run an initial static FEA study to determine the stress profile and initial factor of safety. It will also be used for our first optimization study.



Figure 7: C-clamp model prior to optimization

Now that we have our initial C-clamp model before optimization, we can apply the loads and constraints determined above and provide a visual of them. Figure 8 shows the fixed constraint acting on the top clamping surface of our model. Figure 9 shows the load of magnitude 10,675N acting downward and tangentially to the walls of the extruded cut we created to simplify the model threads. This strategy was obtained from reference [1]. This load represents the clamping force generated when tightening the screw.

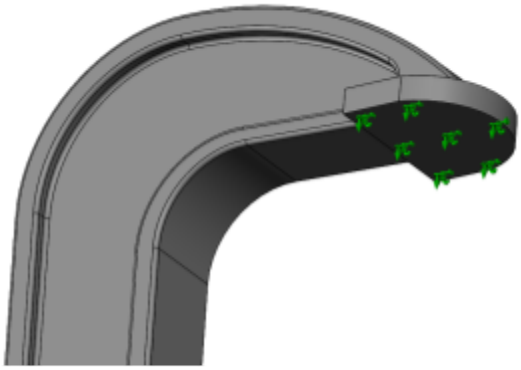


Figure 8: Visual of constraint

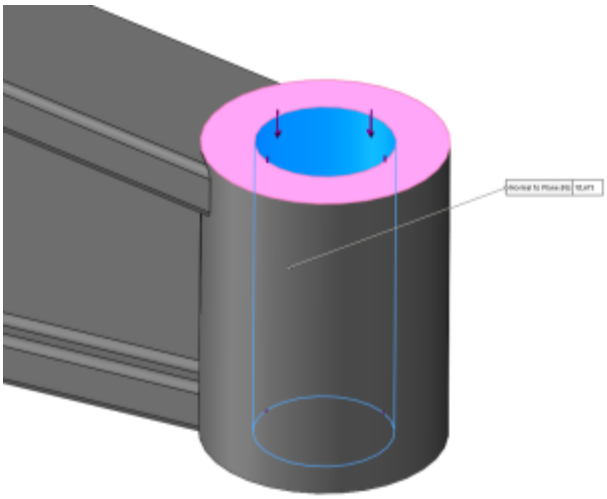


Figure 9: Visual of load

Before we run any simulations, let's be specific and clarify our objectives being minimized and the constraints that we want to satisfy. Our objective is to minimize the mass of the C-clamp from our original model. This will be done by running SolidWorks topology studies to see where material of the C-clamp can be removed. It should also be noted that after using the evaluate tool in SolidWorks, the initial mass of our C-clamp is 2.44 lbs. Now, the constraint we must satisfy is that the factor of safety of our optimized C-clamp model must be very close to the factor of safety of our unoptimized model. This ensures that the optimized clamp is still able to withstand the maximum load rating of our reference clamp with the same FOS. This FOS value will be specified below after an initial static study is run. Now we can proceed with the simulation tools we want to employ.

For our optimization, we decided to use SolidWorks simulations. We will first run a standard static study and then use the SolidWorks Topology simulation add-on to iterate on our C-clamp and optimize its shape. Using this topology simulation will allow us to observe where we can remove material while maintaining the same factor of safety.

In order to run our topology optimization process, we need to first run a standard static study on our initial design to determine the factor of safety of our C-clamp prior to optimization. This is an input parameter for our topology optimization study, so SolidWorks knows how much material can be removed. The visual of the initial design prior to optimization is shown above in Figure 7. The visual showing the summary of loads and constraints for our study is shown below in Figure 10. Finally, the visual of the von Mises stress plot of our initial design is shown in Figure 11.

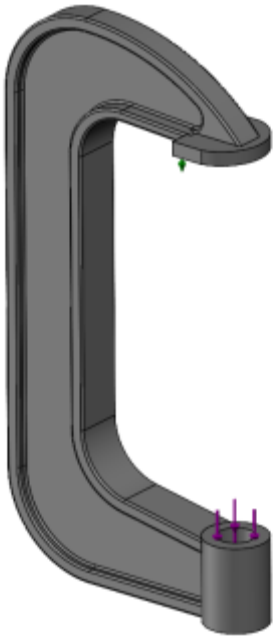


Figure 10: Summary of loads and constraints

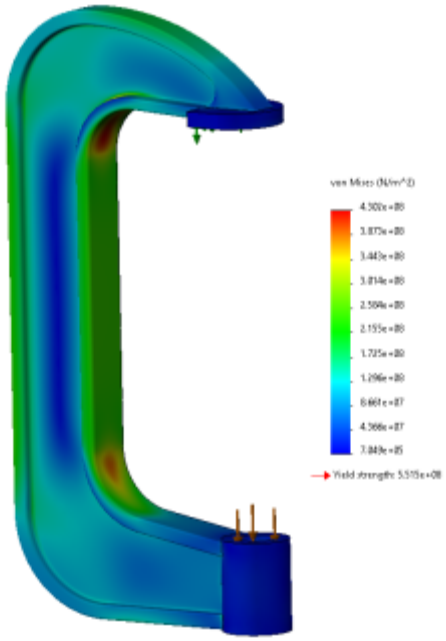


Figure 11: Von Mises stress profile for initial design

From the static study, we can now determine our initial FOS. This simulation allowed us to find that the max stress for the unoptimized model, based on our loads and constraints, is $4.302*10^8$ N/m^2. Using the yield strength of ductile cast iron of $5.515*10^8$ N/m^2, we can find the factor of safety of the unoptimized clamp by calculating yield strength/max stress, which is equal to $(5.515*10^8)/(4.302*10^8) = 1.282$. This means that our constraint to satisfy is that our final optimized C-clamp must have a factor of safety of at least 1.282. The summary of the goals and constraints for our topology study is shown in Figure 12.

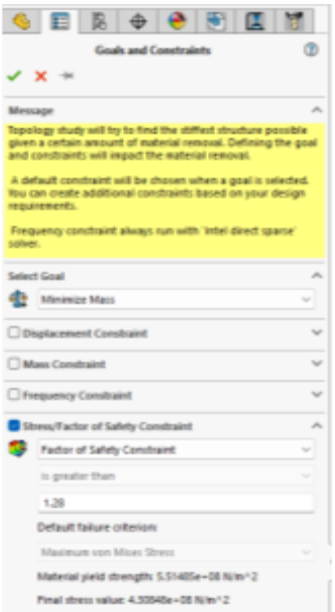


Figure 12: Topology optimization objective and constraint

Results

Now, we are able to proceed with the topology studies in order to iterate and optimize our C-clamp model. After running our first topology optimization, with the material, loads, constraints, and goals and constraints specified in the method, we were given the results shown in Figure 13. This was used to redesign our C-clamp model by removing material that SolidWorks thought was "Ok to remove". Our first iteration redesigned model used a slot along the middle body of the clamp and is shown in Figure 14. Using the evaluation tools, we can see that this first iteration has a mass of 2.19 lbs. Compared to the original mass of 2.44 lbs, we can already see that the mass has been reduced by 0.25 lbs.

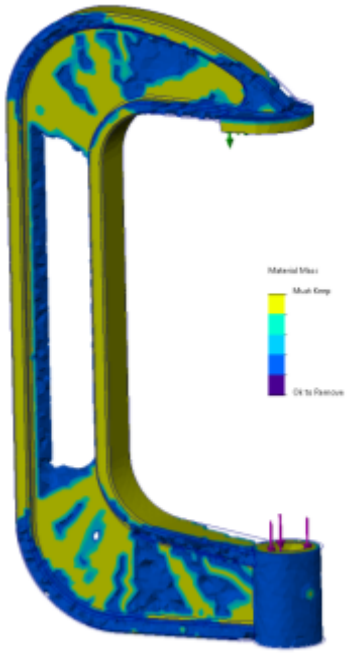


Figure 13: First iteration topology results



Figure 14: First iteration model redesign and mass

After doing a first redesign of our C-clamp model, we then ran the topology optimization study again using the same study settings as the previous study. The results of this study are shown in Figure 15. From the blue and blank areas that can be seen on the top and bottom handles of the clamp, we redesigned the model by removing the area in those sections using an extruded cut. The second iteration design can be seen in Figure 16. As we did previously, using the evaluate tool in SolidWorks, we see the new design has a mass of 2.13 lbs.

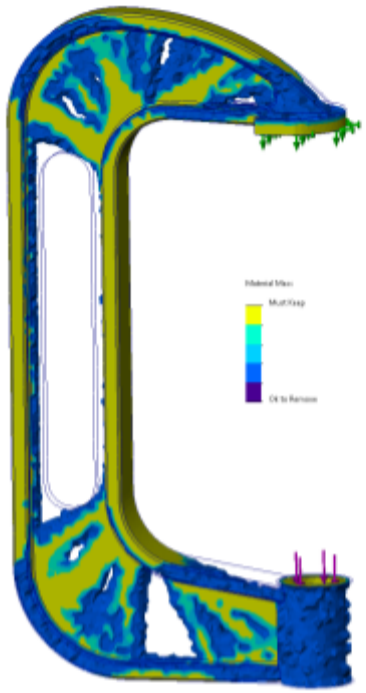


Figure 15: Second iteration topology results

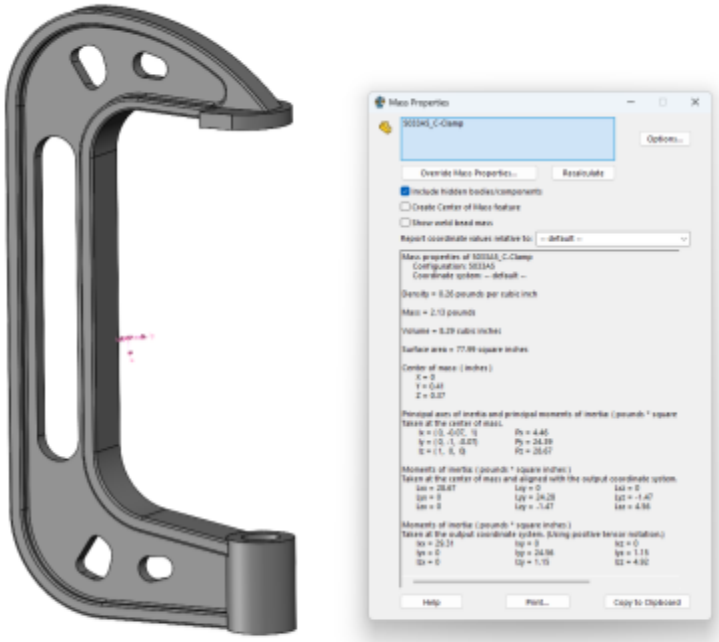


Figure 16: Second iteration model redesign and mass

To be confident that our final design satisfies the constraint that our optimized C-clamp has a factor of safety that is at least that of the original C-clamp, let's run one final static study with the same loads and constraints we have been using throughout this process. This study is shown in Figure 17. Calculating the FOS of the optimized clamp, we find that our FOS is yield strength/max stress = $(5.515*10^8)/(4.149*10^8)$ = 1.329 FOS. This tells us our removal of mass for optimization was successful based on our optimization constraint of keeping the FOS at least 1.282. Therefore, this second iteration redesigned model is our final, optimized C-clamp design.

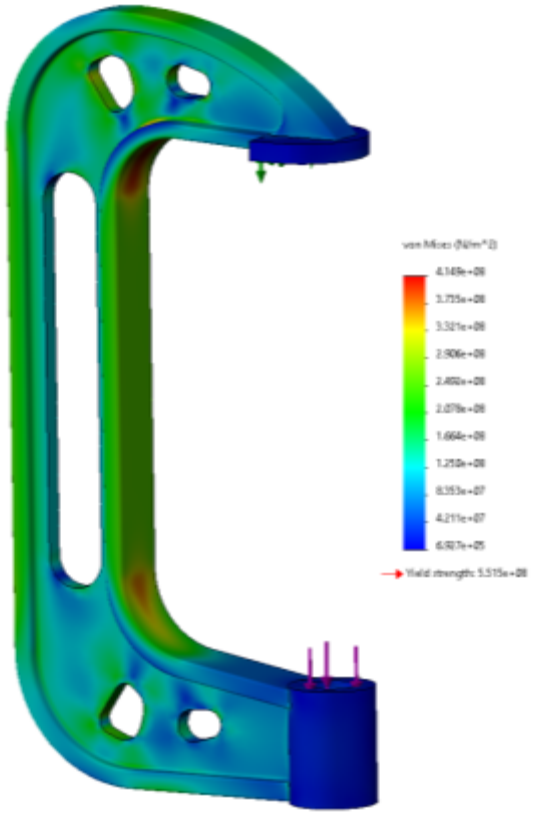


Figure 17: Final static study of optimized clamp model

Now, let's compare our final optimized design versus our initial reference design. A better visual of our final optimized C-clamp design is shown in Figure 18. A visual of our initial C-clamp design, which was altered from the McMaster Carr model is shown in Figure 19.

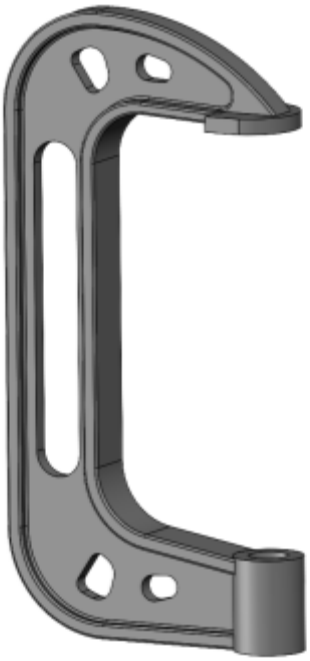


Figure 18: Final optimized c-clamp design

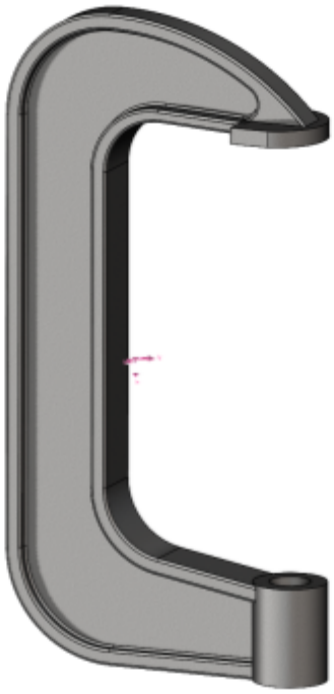


Figure 19: Initial unoptimized design

From visually viewing the two models, you can easily see areas where we removed material from our unoptimized model. The middle of the body has a large slot taken out of it, and the top and bottom surfaces of the clamp also have smaller sections taken out. Let's compare the two designs more quantitatively now. Using the evaluate tool, we found the initial mass of the unoptimized design to be 2.44 lbs and the mass of the optimized design to be 2.13 lbs. Therefore, with our mass optimization by removing material, we were able to reduce the mass by 0.31 lbs. This is a mass reduction of around 12.7% from the initial design, which is a relatively significant mass optimization. From the static FEA studies run above in Figures 11 and 17, we can also quantitatively compare the FOS of each design. As calculated before, the FOS of the initial design was 1.282, and the FOS of our final design is 1.329. We can see that by optimizing our C-clamp, we were able to actually improve the factor of safety by 0.047, or a 4% increase from the initial FOS. Therefore, our final C-clamp design is more optimal as it decreases the mass by 12.7% while increasing the factor of safety by 4%.

After comparing the overall initial and final designs, we can also compared a smaller area of interest for our initial and final designs to further understand our results.



Figure 20: Initial design area of interest mass



Figure 21: Final design area of interest mass

For our specific area of interest comparison, we decided to compare the mass of the section of the C-clamp that the user's hand would grip during use. This is where the most material was removed, so we thought it would be insightful to do so. The visual of the section of comparison can be seen in Figures 20 and 21 above. By using the evaluate function, we found that the mass of the middle section of the initial design was 0.93 lbs, and the mass of the middle section of the final design was 0.69 lbs. This reduction of 0.24 lbs between the initial and final designs of the middle portion of the clamp is quite significant. This reduction of 25.8% signifies that a lot of the mass of this middle portion of the clamp is not actually needed in order for it to support a load without plastically deforming.

Conclusion

During our simulation, our objective was to minimize the mass of the model while maintaining the original FOS. Motivations for this study are straightforward and include removing mass to reduce the cost of production, as well as improving the ergonomics of the tool. From our optimization study, we were able to make extruded cuts throughout the body of the clamp to

remove mass, bringing the total mass from 2.44 lbs down to 2.13 lbs. This resulted in a total mass change of 0.31 lbs, or approximately 12.7% of the original mass. Running a static study on the final model, we found an FOS of 1.329, which is slightly higher than the original FOS of 1.282. This confirms that our final redesigned model not only maintains a comparable FOS to the original model's FOS of 1.282, but also does so with less material, which is exactly what we set out to achieve using the topology optimization tool. Therefore, our FOS constraint outlined throughout this document is satisfied. This slight increase in FOS can be attributed to stress distribution changing from the original model to the final model due to the SolidWorks topology optimization solver's algorithm. The bulk of the material was removed from a low-stress region, meaning the removed material was not significant in contributing to bearing the clamping loads. As a result, the stress distribution should remain largely unchanged, keeping the max stress nearly the same and allowing the FOS to be very similar as well, which is consistent with our results.

After running two iterations of topology optimizations, we realized that the first iteration had a far more significant effect on the design. Moving from our initial model to our first redesign, we were able to remove 0.25 lbs, compared to only 0.06 lbs removed between the first and second redesign. This demonstrates that the bulk of our mass savings came early in our optimization process, while during the second iteration, our mass optimization was focused on smaller, more precise areas for refinement.

Specifically, when changing from the original model to the first redesigned iteration, we removed material on the handle of the C-clamp using a slot and an extruded cut in SolidWorks, which can be seen in Figure 14. This cut was where we lost most of the mass during our optimization study, with approximately 10.25% of the original mass being cut, making it the most significant modification in our entire optimization study. Despite the material removal from the middle section of the clamp, the C-clamp is still able to maintain structural integrity under clamping loads.

There are several directions for future work that can build on this project. First, an FEA study should be conducted to evaluate how the pressure from a hand gripping the handle will affect the handle area of the C-clamp, where a bulk of our material was removed. While removing material in that area maintained the FOS for clamping, the reduced material might limit the handle's ability to withstand grip pressure during use. This should be tested before machining to ensure that the handle doesn't plastically deform during use, since this can pose a safety risk to the user.

Another area for future work would be to run an optimization study using Fusion's shape optimization or generative design tools and compare those results to ours. If the results are similar, it would be a strong validation that our redesigned C-clamp is a sound design. With more independent references pointing to similar conclusions in design, we can have more confidence in our approach to redesigning.

Finally, the most definitive way to validate our optimization is to physically machine the redesigned C-clamp in real life and conduct real-world load testing. By loading the C-clamp to its holding capacity and using a DIC algorithm paired with a high-resolution camera, we can directly observe whether the C-clamp plastically deforms under load. This experiment would bridge the gap between our simulation results and real-world performance, which will give us reliable confirmation that our design is optimized and safe to use.

Works Cited

[1] SolidWorks Tutorial, "SolidWorks tutorial #357: C-clamp with simulation analysis," *YouTube*. [Online]. Available: https://www.youtube.com/watch?v=V8GUvA5tDHs.

[2] ChefOtter2676, "Wk4 elasto-plastic analysis of C-clamp," *CliffsNotes*, Mar. 11, 2024. [Online]. Available: https://www.cliffsnotes.com/study-notes/17055563

[3] McMaster-Carr Supply Company, "5033A5," *McMaster-Carr*. [Online]. Available: https://www.mcmaster.com/5033A5/.